

Algorithms 2

Vertex-connectivity in graphs

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1. Connectivity

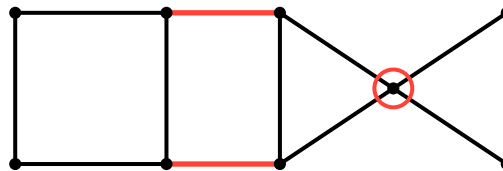
Definition 1.1.1

- For a graph G , write

$$C(G) := \text{number of connected components of } G$$

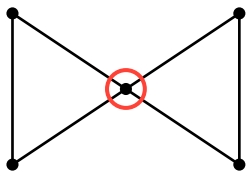
Definition 1.1.2

- Let $X \subseteq V(G) \cap E(G)$.
- If $C(G - X) > C(G)$, then X is called a *disconnector*



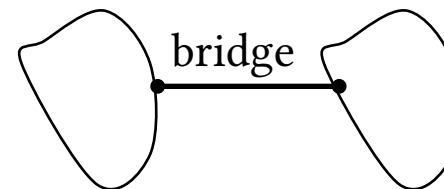
1.1 Graph disconnectors

if $X \subseteq V(G)$, then X is a *vx-disconnector*.



- vx-disconnectors of size 1 are called *cut-vxs*.
- The ends of bridges are cut-vxs.
- The opposite is **not** true!

if $X \subseteq E(G)$, then X is a *edge-disconnector*.



- edge-disconnectors of size 1 are called *bridges*.

Observation 1.1.3 (H.W.)

$e \in E(G)$ is a bridge $\Leftrightarrow e$ does not lie on a cycle of G

2. Whitney's Theorem

Theorem 2.1.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vxs $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

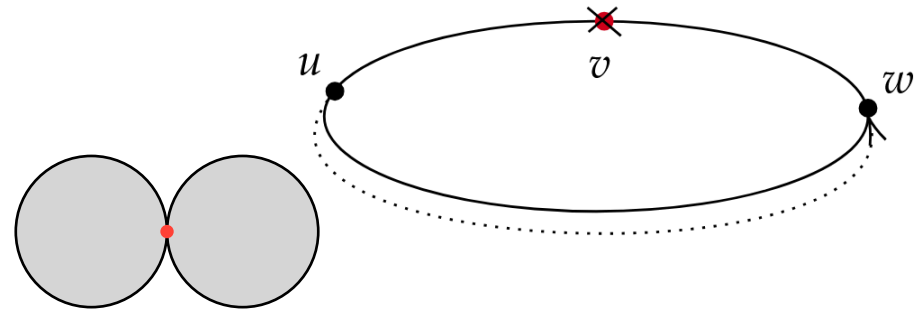
Theorem 2.1.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vertices $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

proof $\Leftarrow$.

- Assume G has a cut vertex, denoted by v .
 - v separates the graph into 2 components C_1, C_2 .
 - Take $w \in C_1, u \in C_2$ both should hold true
1. A cycle $C := wx_1 \dots x_\ell uw$ exists in G .
 2. If we remove v then u and v are disconnected.



contridiction: If $v \in C$ then removing v or any vertex in C wont separate w from u , and if $v \notin C$ then the path $wx_1 \dots x_\ell u$ still exists.

Theorem 2.1.2 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vxs $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

$\Rightarrow$

- Suppose G has no cut-vxs.
- We prove by induction on

$\delta_{G(u,v)} :=$ length of the shortest path $u \rightsquigarrow v$ in G

- Basis:
 - Let $u, v \in V(G)$ have $\delta_G(u, v) = 1$
 - If $uv \in E(G)$ is a bridge, the u, v are cut-vxs. **contradiction.**
 - From the observation uv lies on a cycle.

Observation 2.1.3 (H.W.)

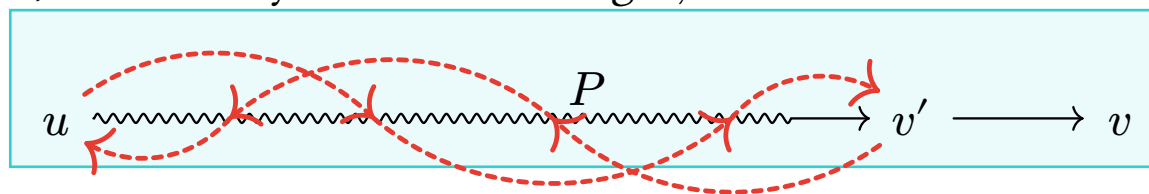
$e \in E(G)$ is a bridge $\Leftrightarrow e$ does not lie on a cycle of G

Theorem 2.1.4 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vertices $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

- Suppose G has no cut-vertices.
- Suppose that the claim hold true for all pairs of vertices u, v such that $\delta_{G(u,v)} < k$
- Consider a pair $u, v \in V(G)$ satisfying $\delta_G(u, v) = k$
- Let P be the shortest path $u \rightsquigarrow v$, and let v' denote the vertex before v on P .
- As $\delta_G(u, v') = k - 1$, there is a cycle C containing u, v'



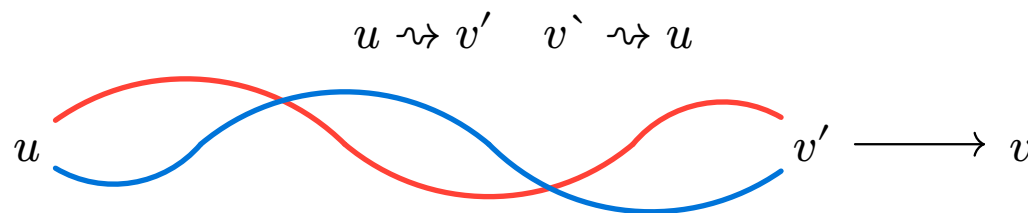
- If $v \in V(C)$ we are done

Theorem 2.1.5 (Whitney's Theorem)

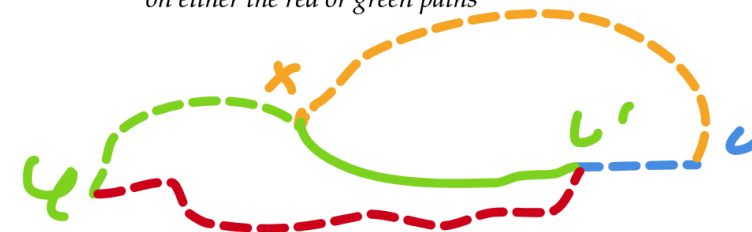
- G - a connected graph with $v(G) \geq 3$

G has no cut-vertices $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

- Otherwise, we can divide C into 2 arcs



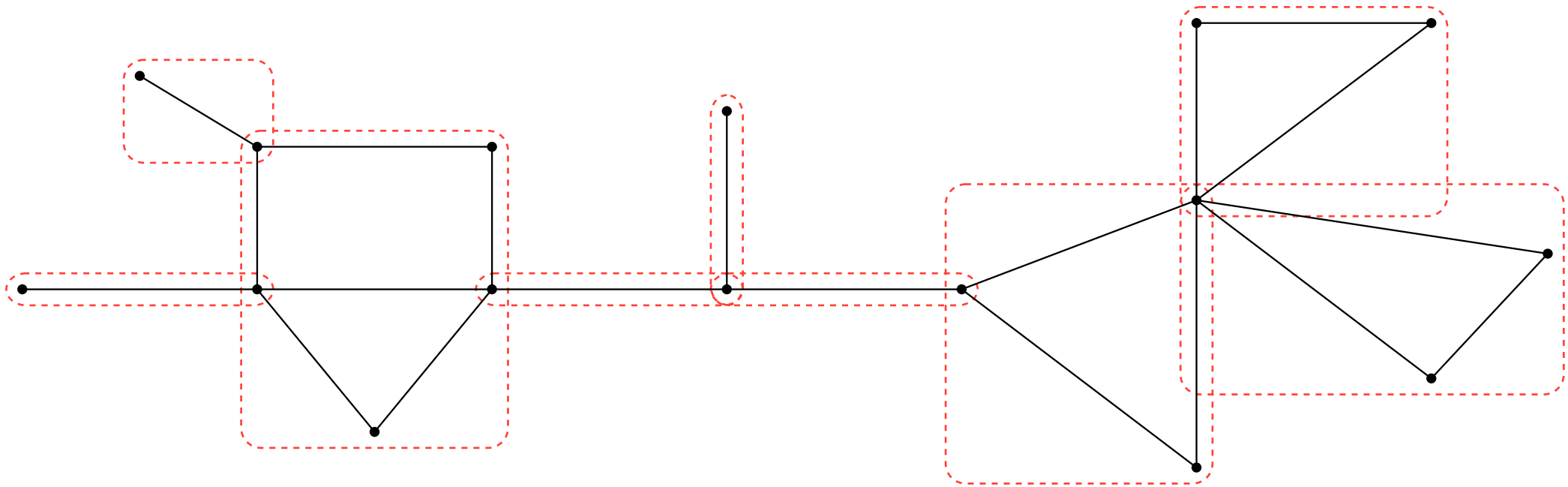
x is the first vertex v meets
on either the red or green paths



- As G has no cut-vertices, $G - v'$ is still connected,
- There is a path from v to a vertex x that lies on one of the arcs)
- W.L.O.G, let $x \in u \rightsquigarrow v'$ **take the shortest path from v to any of the arcs!**
- Then the cycle $u \rightsquigarrow x \rightsquigarrow v \rightarrow v' \rightsquigarrow u$ is a cycle, and contains both u and v .

3. Consequences of Whitney's Theorem

Definition 3.1.1 A maximal connected subgraph of G containing no cut-vertex is called a block of G



Theorem 3.2.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

G has no cut-vxs $\Leftrightarrow$ any 2 vertices of G lies on a common cycle.

- Any two blocks of G meet in at most one vertex.
- If two block do meet, their intersection vertex is a cut-vertex.
- The blocks of G partition $E(G)$ (H.W.).
- Every cycle of G lies in precisely one block of G (H.W.).

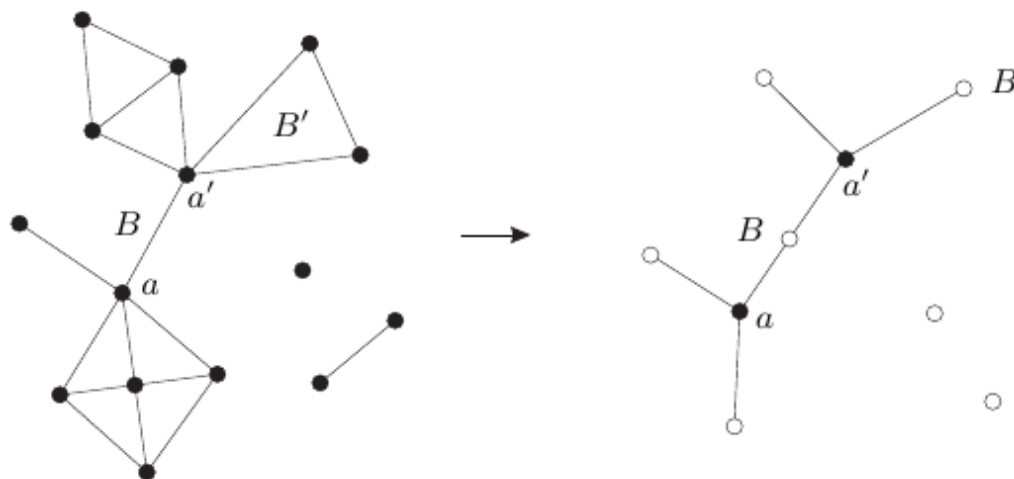
Given a graph G define the following auxiliary graph:

- $B(G) := \{B : B \subseteq G \text{ is a block of } G\}$
- $C(G) := \{v : v \in V(G) \text{ is a cut-vertex of } G\}$

Let $BC(G)$ denote the following graph

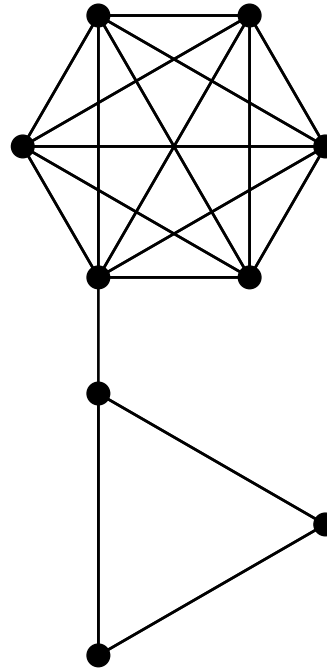
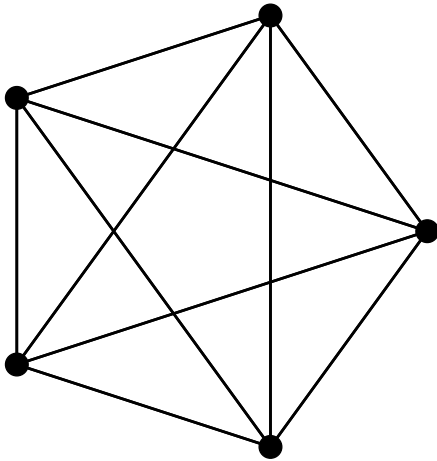
- The vertices of $BC(G)$ are $B(G) \cap C(G)$
- The edges of $BC(G)$ are $\{\{B, v\} : B \in B(G), v \in C(G) \text{ and } v \in B\}$

Theorem 3.3.1 if G is connected then $BC(G)$ is a tree

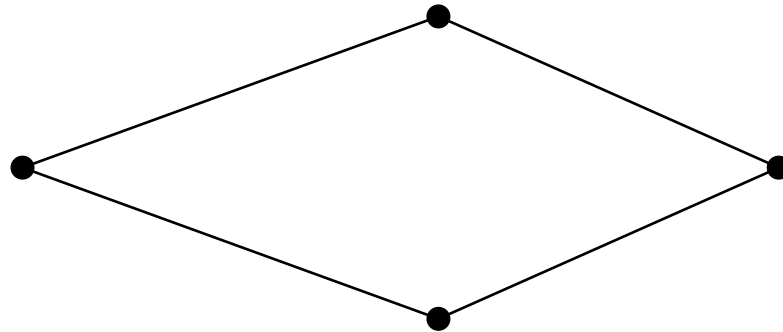


4. *Assesing connectivity*

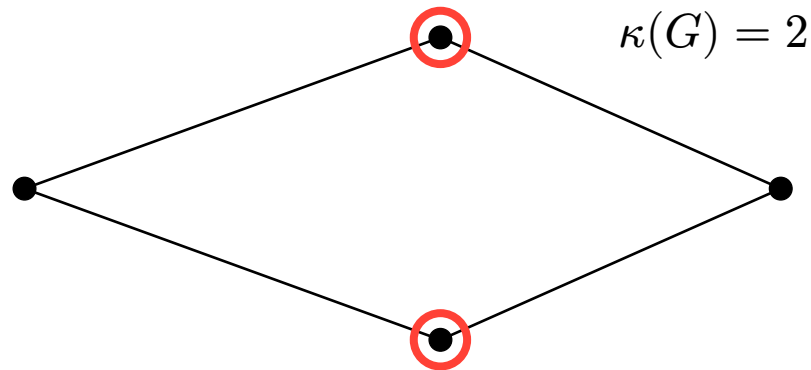
Question 4.1.1 Given these graphs, which one is more connected?



Definition 4.2.1 The minimum size of a vx set X such that $G - X$ is disconnected or has a single vx is called the *vx-connectivity* of G and is denoted by $\kappa(G)$



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Remark

Other than the complete graph:

A graph is k -connected $\Leftrightarrow$ all of its vx-cuts are of size $\geq k$

The complete graph has no vx-cuts of any size, this is where the 2nd part of the definition comes in

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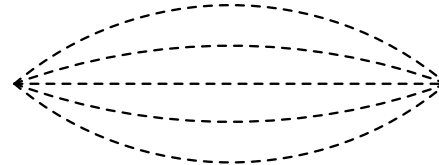
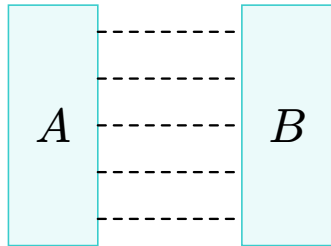
If G is k -connected then G is also $k - 1, k - 2, k - 3, \dots, 1$ -connected.

But it can also be $k + 1$ connected, we don't know if it is!

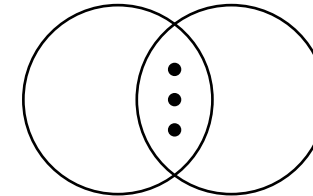
All we know that any $k - 1$ vertices removed from G won't disconnect it.

Definition 4.4.1

- G a graph
- $A, B \subseteq V(G)$
- let $\rho_G(A, B) :=$ the number of vx-disjoint (A, B) -paths



Here $|A| = |B| = 1$



$A \cap B \neq \emptyset$ also can happen

Remark

Vertices in A and B counts toward the vertices of the paths.

If $|A| = 1$ or $|B| = 1$ then we count the number of paths from the neighbours of A or B respectively.

Definition 4.5.1 G is *k -connected* if $\rho_G(u, v) \geq k \quad \forall \{u, v\} \in \binom{V(G)}{2}$

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So whats better: vertex connectivity or path connectivity ?

Menger's theorem tells us that they are the same!

5. Menger's theorem

Definition 5.1.1

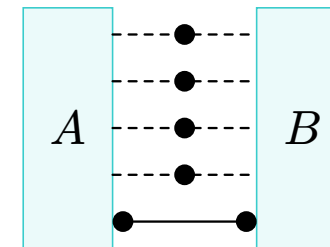
- $A, B \subseteq V(G)$
- Denote by $\kappa_G(A, B)$ the size of the **least** (A,B)-vx-disconnetor

Theorem 5.1.2 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

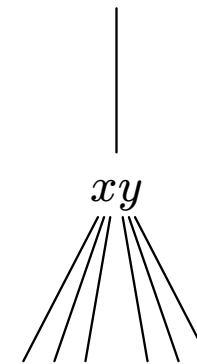
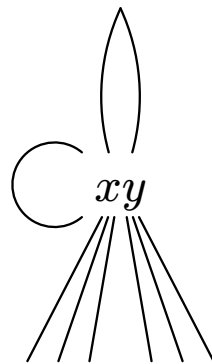
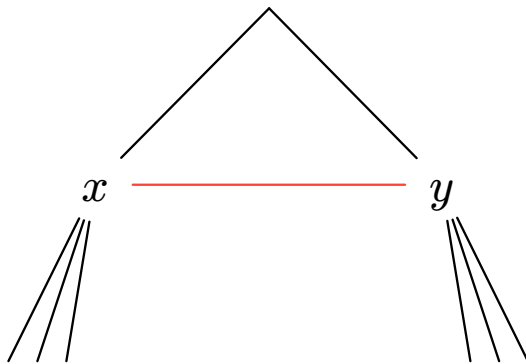
$$\kappa_G(A, B) = \rho_G(A, B)$$

- You know this by the name “min-cut-max-flow”
- $\kappa_G(A, B) \geq \rho_G(A, B)$ is trivial
- We need to prove

$$\kappa_G(A, B) \leq \rho_G(A, B)$$



- Consider the edge $e = xy$
- To contract e :
 - remove e
 - merge x and y into one vertex v_{xy}
 - connect v_{xy} to $N_G(x) \cap N_G(y)$, remove any duplicant edges that may arise



- Denote by G/e the graph resulting from contracting e

Theorem 5.3.1 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\kappa_G(A, B) = \rho_G(A, B)$$

- Set $\kappa := \kappa_G(A, B)$ and $\rho := \rho_G(A, B)$
- Proof by induction on $e(G)$
- If $e(G) = 0$ the graph is disconnected
 - $\rho = |A \cap B| = \kappa$

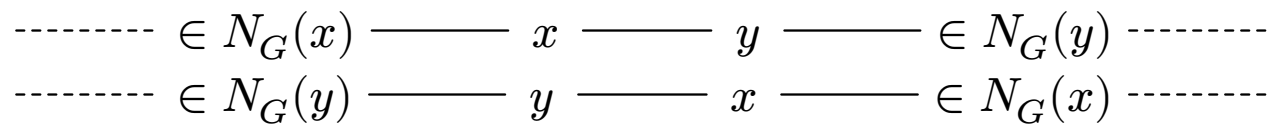
Theorem 5.3.2 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\kappa_G(A, B) = \rho_G(A, B)$$

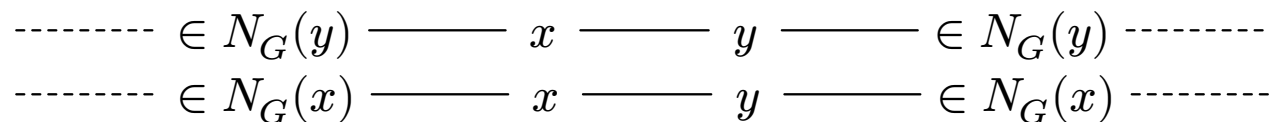
- assume $e(G) > 0$
- let $e = xy \in E(G)$ be an arbitrary edge
- In G/e , define $A_e \subseteq V(G/e)$ as follows
 - If $x, y \notin A$, then $A_e := A$
 - if $x \in A, y \notin A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
 - if $x \notin A, y \in A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
 - if $x, y \in A$, then $A_e := (A \setminus \{x, y\}) \cup \{V_e\}$
- Define B_e in the same manner

- Suppose that G/e has κ vx-disjoint (A_e, B_e) -paths
 - Any path not containing v_e exists in G
 - if none of the paths here contains v_e , then we are done
- suppose that there is a path in the linkage containing v_e .

xy can be expanded in the path

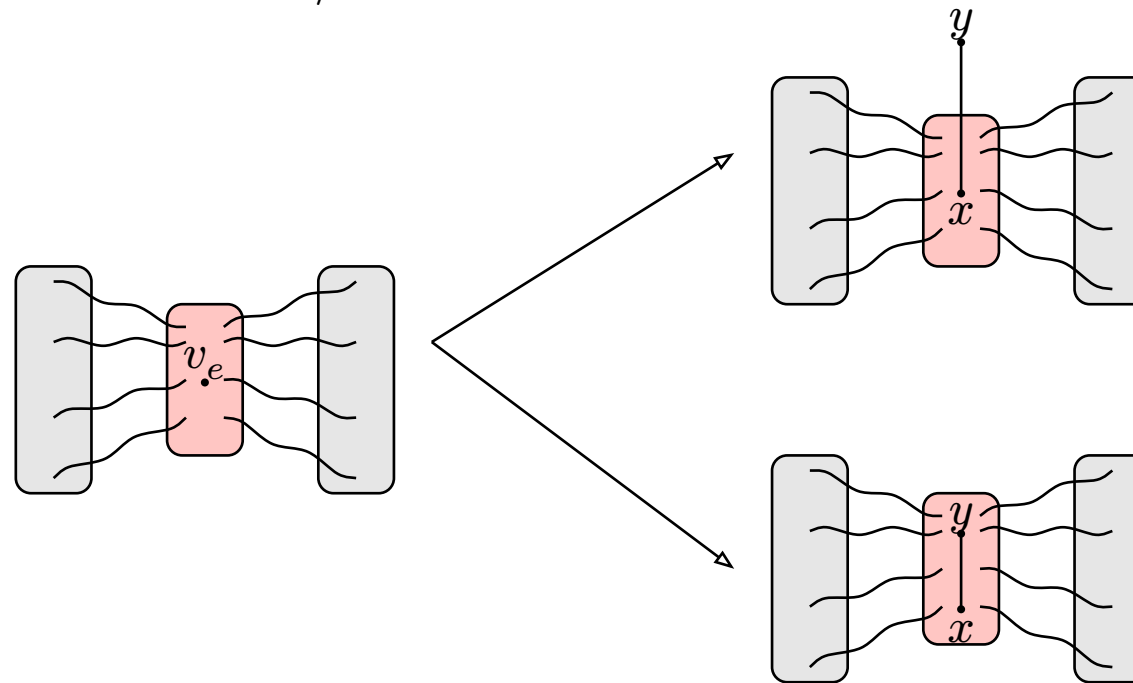


xy cannot be expanded in the path



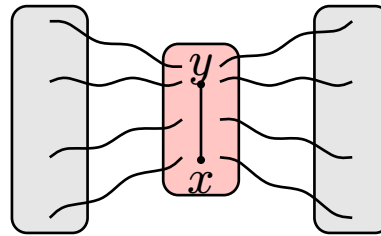
- In the second case, just take one of the verticies.

- Suppose that G/e don't have κ vx-disjoint (A_e, B_e) -paths
- $\kappa_{G/e}(A, B) < \kappa$
- Any (A_e, B_e) -vx-disconnector in G/e contains v_e
 - if not, G has an (A, B) -vx-disconnector of size $< \kappa$
- Any (A_e, B_e) -vx-disconnector in G/e is of size $\kappa - 1$



We are now in this position:

- $X := (A, B)$ -vx-disconnector (Maybe $X = A$ or $X = B$)
- $|X| = \kappa$
- X spans e



- Note that by I.H.

$$\kappa_{G-e}(A, X) \geq \kappa \quad \text{and} \quad \kappa_{G-e}(X, B) \geq \kappa$$

Corollary 5.5.0.1 (H.W.)

- Let G be a graph. Then

$$\kappa(G) = \min\{\rho_G(u, v) : u, v \in V(G), u \neq v\}$$

Lemma 5.5.1 (The Extension Lemma H.W.)

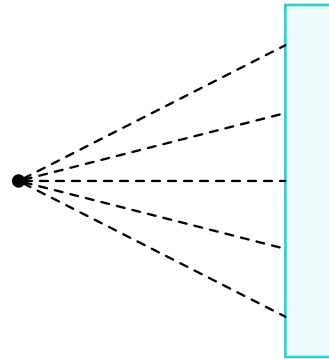
- G - k -connected graph
- H obtained from G by:
 - Adding a new vertex
 - Connecting the new vertex to k arbitrary vertices
- Then H is k -connected,

- Given $x \in V(G)$ and $B \subseteq V(G)$ s.t. $x \notin B$, we write (x, B) -fan to denote:

Lemma 5.5.2 (Fan Lemma H.W)

- G - k -connected graph
- $x \in V(G)$ and $B \subseteq V(G) \setminus \{x\}$

Then G has an (x, B) -fan if size $\min\{k, |B|\}$



Theorem 5.6.1 (Dirac's Theorem)

- $2 \leq k \in \mathbb{N}$
- G - k -connected graph
- $S \subseteq V(G)$ such that $2 \leq |S| \leq k$

Then G contains a cycle constains S

- By induction on k
- for $k = 2$, the claim hols by whitney's theorem.

Theorem 5.6.1 (Dirac's Theorem)

- $2 \leq k \in \mathbb{N}$
- G - k -connected graph
- $S \subseteq V(G)$ such that $2 \leq |S| \leq k$

Then G contains a cycle constains S

- assume $k \geq 3$
- let $x \in S$ be arbitrary and set $T := S \setminus \{x\}$
- $\kappa(G - x) \geq k - 1$ So there is a cycle C in $G - x$ containing T .
- let F be an (x, C) -fan in G of size $\min\{k, v(C)\}$
- Note the T partitions C into $k - 1$ arcs
- At least two of the paths in the fans land in the same arc on C
- We can extend C to contain x